

Abdullah Shamaail

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SUMMARY

Senior AI Software Engineer (PhD Candidate) with 5+ years of experience transforming advanced machine learning research into high-fidelity production infrastructure. Expert in designing deep learning architectures for high-dimensional geospatial and spatio-temporal datasets utilizing Python and modern C++. Strong mathematical foundations in linear algebra, geometry, and numerical optimization, delivering up to **65× computational speedups** in remote Unix environments.

EDUCATION

Iowa State University <i>Ph.D. Computer Engineering (GPA: 3.90/4.00)</i> • Focus: Scalable Spatio-Temporal Data Management and pattern mining in distributed environments. • Honors: NSF Student Travel Grant Recipient; Preparing Future Faculty Associate (2025).	Ames, IA <i>Expected Aug 2026</i>
Iowa State University <i>M.S. Computer Engineering (GPA: 3.89/4.00)</i>	Ames, IA <i>Dec 2024</i>
Lahore University of Management Sciences <i>B.S. Computer Science</i>	Lahore, Pakistan <i>May 2020</i>

EXPERIENCE

Iowa State University <i>Graduate Research Assistant</i> • Algorithm Implementation: Designed the SDC framework using transformers, indexing, and spatial partitioning, achieving up to 65× speedups in diversity-aware clustering for dynamic, large-scale geospatial and spatio-temporal datasets. <i>IEEE MDM 2026</i>. • Mathematical Optimization: Engineered BARMC, a high-performance analysis framework that achieved an 80% speedup in hydrogen-bond detection by implementing a loss-less spatial pruning algorithm that pruned 90% redundant data points utilizing strict linear algebra and numerical methods. <i>ACM SIGSPATIAL 2024 and TSAS 2026 (Extension)</i>. • High-Dimensional Data: Developed a domain-specific trajectory compression algorithm that achieved an 80% reduction in storage and a 3× speedup in data retrieval for MD simulations. <i>ADBIS 2022 and Information Systems 2024 (Extension)</i>. • Data Infrastructure: Optimized large-scale text analytics using Hadoop MapReduce, resulting in a 2× increase in processing throughput for bigram frequency identification within remote Unix system clusters.	Ames, IA <i>Jan 2021 – Present</i>
Altair Engineering Inc. (A Siemens Company) <i>Engineering Data Science Intern</i> • Applied Research: Engineered GNN and Transformer-based architectures for PhysicsAI, achieving a 10× reduction in MAE and enabling longer-horizon predictions for large-scale physical simulations. • Scaling & Deployment: Deployed scalable ML pipelines using PyTorch and TensorFlow, supporting end-to-end training and GPU-accelerated inference integrated with production platforms.	Troy, MI <i>May 2025 – Aug 2025</i>

TECHNICAL PROJECTS

AutoJobScout – Agentic AI & RAG Platform <i>LangGraph, RAG, LLMs, Python</i> • Architected a multi-agent workflow using LangGraph, achieving 92% matching accuracy via semantic resume analysis and resume-to-JD alignment. • Optimized asynchronous microservices to handle 1,000+ daily listings, reducing end-to-end inference latency to <15 seconds on Unix-hosted systems.	2025
Resume Tailor – AI-Powered ATS Optimizer • Developed a client-side web application leveraging Groq’s LLaMA API for dynamic resume tailoring, implementing intelligent keyword optimization and content reordering based on job descriptions.	2025
F1 Telemetry Dashboard <i>C#, Blazor WASM, Python, Azure</i> • Built a full-stack dashboard to visualize high-velocity streaming numeric telemetry, utilizing Blazor WebAssembly for high-performance client-side rendering with zero server latency.	2025
Sarcasm Detection Engine <i>LSTM, SVM, Python</i> • Trained a deep learning model for news headline classification, achieving 84% precision and 82% recall using tokenized dense vector representations and non-linear optimization.	2022

TECHNICAL SKILLS

Machine Learning & AI: Computer Vision, Deep Learning, Linear Algebra, Geometry, Numerical Optimization, NLP (Transformers), GNNs, CNNs, LSTMs, RAG, Agentic Workflows.
Big Data & Infrastructure: Unix Shell (Remote Deployment), PySpark, Hadoop MapReduce, Apache Flink, Pig Latin, SQL, NoSQL, Git, Linux, GPU Inference.
Frameworks & Tools: PyTorch, TensorFlow, Keras, Scikit-learn, MLflow, LangChain, Docker, Azure, Databricks.
Programming: Python (Expert), C++ (Prior experience interfacing pipelines), Java, JavaScript, MATLAB, C# (Blazor), SQL.